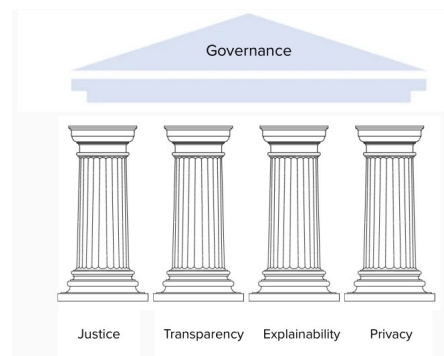


The McAllister AI Ethical Framework for Banking

Michael McAllister
101359469



Ethics in artificial intelligence has been a rising concern in recent years, in fact multiple scandals have resulted from deployed AI models¹. Artificial intelligence has moved from science fiction to reality, and this has caused fear about its potential misuse. The best way to allay people's fears is to work within an ethical framework, this will ensure that your model will be adopted by stakeholders and help ensure a model's success¹. Each principal for the ethical framework will be provided in two parts, a guiding principal in an easy to ingest statement that should be considered throughout the entire process, and an in-depth description of the principal. I present to you the McAllister AI Ethical Framework, based on the 5 principals of Justice, Transparency, Explainability, Privacy and Governance.

Justice

Guiding Principal: Be fair to everyone who will come into contact with your model

For an AI model to be just it must promote fair treatment, promote equitable outcomes, attempt to be as inclusive as possible, and be as unbiased as possible. '85% of AI projects through 2022 will deliver erroneous outcomes due to bias in data, algorithms or development teams'². The concept of non-maleficence falls under justice as well, an AI model should not harm any individuals or present an existential threat to humanity. Justice is not only an ethical issue, but also an adoption issue and success issue, without trust in your AI model people will not use it. Harm includes any discrimination due to bias, where harming other individuals means an AI model should be ended.

Transparency

Guiding Principal: Every step of the model must be tracked, codified and reviewed, and be available for those who're impacted by the model.

AI needs to follow a similar methodology to research papers, 'researchers have an ethical obligation to facilitate the evaluation of their evidence-based knowledge claims' by codifying the evidence, analysis, and research design procedures. This also includes being able to provide this material to any parties effected by the AI who request this information.

Explainability

Guiding Principal: You must be able to explain, in detail, why the model is acting how it does

An AI developer must be able to explain why an AI model makes the decisions it makes, this is called prediction accuracy. The people effected by the decisions of the model have a right to know why the decisions made were made, and that they did not suffer from discrimination. This is now a growing field and with current technology this is possible, and the technology will improve over time. Using the best tools and methods is an ethical imperative as technology progresses.

Privacy

Guiding Principal: All personal information must be kept secure, and people's privacy rights must be respected

People have a right to privacy of their personal information. An AI model, and the data used by the model, must be collected and used in a secure matter and it must not be shared. Best practices are to use the strictest international privacy laws as a minimum standard for privacy and data collection/usage methods.

Governance

Guiding Principal: KPIs must be tracked in regards to ethics, with a scorecard provided on the current progress

The final part to any ethical AI framework is a governance framework. A governance framework reduces risk and enables safer and faster growth. Key performance indicators must be set and tracked, along with oversight to confirm compliance with all pillars of the ethical framework.

The McAllister Ethical Framework Applied to Banking, Loan Approvals

The McAllister ethical framework use case is the banking industry, as a loan approval tool for an example. While this framework is robust, the Canada's Federal Government AI Framework³ should also be reviewed as an example of the government's expectations on deploying AI models in Canada. Applying an ethical framework will ensure that a model is being deployed ethically, which in turn ensures success and will help adoption.

The first step in any modeling is defining the goal and application of the model to see if the model can meet the needs of all stakeholders. Do we have the data we require to make an accurate prediction, and is the data biased in any way? What is it we're trying to do, and can this be done in an ethical way? What ethical concerns need to be top of mind during our design phase? A good example is a loan approval model inherently discriminates against people by their ability to repay a loan, is this ethical? Is discriminating against people by their ability to repay a loan just a way to add complexity on top of the real goal of discriminating against the poor, and actually a method of enforcing class hierarchy? These questions are vital, and by asking them at the beginning it will help not waste resources on an AI project that will not be deployed.

In addition, every step of the process needs to be codified and tracked for governance purposes, discussed more later on in the framework.

If the decision is made to proceed then the next step is to ensure that your dataset isn't biased which falls under the Justice pillar. A review of the legislation where your model will be deployed is key, as certain groups have anti-discrimination laws in place, so if your model discriminates against these groups it is breaking the law. The best method to ensure no discrimination against protected groups is to stratify your data to ensure the model will not have a lower loan approval rate against any protected group.

The next ethical decision you need to make is whether you should stratify your data along other lines, and whether these lines are encouraging justice or not. Stratifying different groups to ensure their accuracy is representative of your overall model is great, but you must be careful about why you're choosing one group and not another. Is your own bias at play? Historically marginalized groups have a right to not be discriminated against, but how can you be sure you're not adding bias by adding some groups and not others? A great example would be ensuring that your model does not discriminate against left handed people would not add value, and are you only adding this to your model because you're left handed?

Another major concern is privacy of your data. Best practices are to screen/clean your data and to make it as accurate as possible because your model will have a real impact on people's life, you have an moral obligation to make the model as accurate as possible. Then mask the data so the people developing the model do not have access to people's personal information, data masking ensures that accuracy is not lost while also protecting people's private information. Data masking also prevents hackers from having access to people's personal information in case of a data breach. As a best practice, it is best to automate data screening and masking to attempt to limit human contact with people's personal information as much as possible.

Once the data is cleaned/screened and masked it's time to create a model, and once the model is complete the next step in the process is explainability. Explainability for neural networks is a field that's being researched intensively and advancing quickly. The best current methods to explain the model's decision are open source, available on IBM⁴ and Google's⁵ AI ethical framework page. Their GitHubs are open source and apply decision trees on the data to try and determine how a neural network makes it's decisions, as well as providing bias metrics such as monotonicity. Another part of transparency is documenting every step of the process so the results can be recreated and reviewed, enforcing transparency around all procedures and decisions made.

Checking your model for bias is done throughout the model's life-cycle, but there are three main areas of focus. The first is ensuring your data is not biased, done during the cleaning process. The second is after you have created a model to ensure that your model is not biased. The third is once the model is deployed, tracking the ongoing prediction accuracy to make sure the model keeps performing well in a live environment.

The final piece of the framework is Governance. Governance is not a pillar in our framework, but rather the structure, systems and practices involved in enforcing adherence to the principals of the framework. For governance to be effective there must be KPIs (key performance indicators) that are tracked over time to ensure compliance with the ethical framework. KPIs are business specific, but an example would be a scorecard from 1-5 on adherence to all pillars, with a failing grade being a show

stopper or pulling it from development or production.

Another important issue is ongoing tracking of the model. An AI model cannot be deployed and released into the world, like Frankenstein, without monitoring and maintenance. Tracking model drift and ongoing A/B testing are required to be done in an ethical manner. AI models should not be made and deployed without maintenance, A/B testing is required to ensure that updates perform better than prior version and to track model drift.

The final piece of governance that must be done as a part of the framework are audits. One individual or leader only has a keyhole view on any issue, and third party reviews are indispensable to determine that you are in compliance with the ethical framework. Bringing in a third party auditing company to review your results is the best method to ensure compliance. Some models are created by an individual person, so inclusive viewpoints are not possible, and it's especially important in these situations to bring in diverse viewpoints from governance and auditing companies.

Once all of these steps have been done help ensure a model has been deployed ethically using an ethical framework, you have done all that you can to ensure it's success. Ethics are an imperative throughout the entire process, and there is much testing that must be done to try and remove any and all bias. While non-maleficence is a great goal, the best goal is to create a model that helps better the world. If AI is the future, then incorporating an ethical framework onto your AI, you're ensuring an ethical future for everyone.

1 - Floridi, L., & Cowls, J. (2019). A Unified Framework of Five Principles for AI in Society. Harvard Data Science Review, 1(1). <https://doi.org/10.1162/99608f92.8cd550d1>

2 – Meulen, R. & McCall, T. (2018). Gartner says Nearly Half of CEOs are Planning to Deploy Artificial Intelligence, 3(2). <https://www.gartner.com/en/newsroom/press-releases/2018-02-13-gartner-says-nearly-half-of-cios-are-planning-to-deploy-artificial-intelligence>

3- Government of Canada (2021). Responsible Use of Artificial Intelligence. <https://www.canada.ca/en/government/system/digital-government/digital-government-innovations/responsible-use-ai.html>

4 – IBM Research. Trusted AI. https://research.ibm.com/teams/trusted-ai?cm_mc_uid=20164137470816096157841&cm_mc_sid_50200000=24555521609615784119

5 – Weinberger, D. Google Pair. Playing with AI Fairness. <https://pair-code.github.io/what-if-tool/ai-fairness.html>